

Khwaish Agarwala

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Skills

Language Skills: Python, C++, C, Java, SQL, JavaScript

Web Development: React, Node.js, Express.js, AngularJS, HTML, CSS, Bootstrap

Big data & Cloud: Hadoop, Spark, Hive, Amazon Web Services

AI / ML: Machine Learning, Generative AI, LLMs, LangGraph, Scikit-learn

Databases: MongoDB, MySQL, SQLite, Supabase, Prisma, PostgreSQL

Tools & Concepts: Git, GitHub, REST APIs, Data Structures & Algorithms, OOP

Experience

Threds | *Software Development Intern*

May 2026 – Jul 2026

- Delivered **10+ production feature enhancements and bug fixes** across frontend and backend CRM modules.
- Refactored reusable UI components across **7+ modules**, improving maintainability and user experience.
- Added dynamic field support, optimized backend logic, and contributed through Git-based Agile development.

Tech Stack: React, TypeScript, Vite, TanStack Query, NestJS, Prisma, PostgreSQL, Better Auth, Git, GitHub environment.

Projects

DriftWatch: **AI-Powered Behavioural Anomaly Detection for Cybersecurity**

[GitHub](#) [Live Demo](#)

- Built a **semi-supervised anomaly detection pipeline** over **121K access logs**, combining a **GRU Autoencoder**, **Isolation Forest**, and **XGBoost** to detect **7 cyberattack types** without labelled training data.
- Reduced **false positives by 37%** by leveraging **SHAP-based explainability** to refine anomaly detection decisions and deployed a **real-time Streamlit dashboard** backed by PostgreSQL for behavioural anomaly monitoring.

Tech Stack: Python, PyTorch, XGBoost, Scikit-learn, SHAP, Streamlit, PostgreSQL

SafeTrail: **Your AI-Powered Travel Safety Companion**

[GitHub](#)

- Developed an **AI-powered travel safety platform** that analyzes geospatial location data using **K-Means clustering** to identify high-risk areas.
- Integrated **Google Maps API** and **Google Directions API** to dynamically generate optimized navigation routes and visualize safety zones on interactive maps.

Tech Stack: Next.js, React, Tailwind CSS, TypeScript, GoogleAuth, MongoDB, Google Maps API, Google Directions API, Scikit-learn, Python

Synapse: **AI-Driven Delivery Issue Resolution Platform**

[GitHub](#)

- Designed an **intelligent logistics monitoring system** to analyze delivery data and customer complaints for detecting **3+ operational issues** including delays, route deviations, and failed deliveries.
- Implemented **LLM-based multi-step workflows using LangGraph** to automate issue classification and generate response suggestions for delivery resolution.

Tech Stack: Python, LangGraph, LLM APIs, HTML, CSS, JavaScript

Education

Vellore Institute of Technology (VIT)

B.Tech. in Information Technology

Aug 2023 – June 2027

CGPA: 8.97

Atreyee DAV Public School

CBSE Class 12th Board

April 2023

Percentage: 94%

Atreyee DAV Public School

CBSE Class 10th Board

April 2021

Percentage: 95.6%

Leadership

Vice Chairperson, IEEE TEMS VIT

Dec 2025 - Present

- Oversee operations and strategic planning for a **100+ member student chapter**, coordinating cross-domain initiatives across technical, editorial, and outreach teams.
- Lead and collaborate with **4+ functional teams** to organize **5+ technical events and workshops**, improving student engagement and participation.

AI/ML & Editorial Mentor, IEEE TEMS VIT

Jan 2025 - Dec 2025

- Mentored **30+ students** on AI/ML fundamentals through projects, technical sessions, and hands-on workshops.
- Led editorial and marketing initiatives, producing **10+ promotional and technical content pieces** for IEEE TEMS events, newsletters, and social media outreach.

Certifications & Achievements

GenAI, SmartBridge

[View Certificate](#)

Hands-on internship program exploring generative AI, prompt engineering, large language models (LLMs), and ethical AI practices using real-world tools certified from Google Developers.